

Sri Lanka Institute of Information Technology

Faculty of Computing

SE2012 - Object Oriented Analysis and Design

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Year 02 and Semester 01

How do we develop an OO Program ?



We look at the problem that needs to be solved.

This is the building we want to build.

In a real world scenario, this could be a Student Information System (SIS)

Identifying Objects needed



These are the Blocks (Objects) that we need.

e.g. In SIS objects could be details of OOC, IWT, students called Manoj, Gayani

How do you create these Blocks (Objects)?



What if we needed to manufacture these blocks. How could we do this?
What do we need to make first?

A Mould (Class)



Once we make a Mould (Class) we can make as many blocks (Object) that we need.

How to group these Blocks (Objects)?



How can we group these Blocks?

We could do it by Shape



A Square Mould (class)

A Rectangle Mould
(class)



e.g. In SIS it could be Student Class, Subject Class

Creating Blocks (Objects) from Moulds (Classes)



A Square Mould (class)



A Rectangle Mould (class)



Blocks (Objects) made

Final Product – Assembling Blocks (Objects)



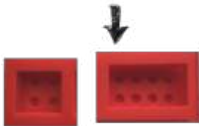
We can now assemble the Objects and create our final solution.



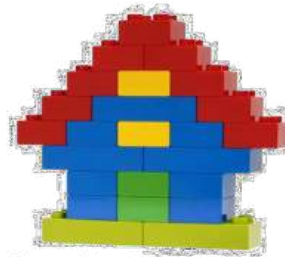
Problem to Solve



Identify Objects that are needed



Identify Classes through Abstraction



Assemble Objects to create the solution



Create Objects from Classes

I am dreaming of





Meeting an Architect to make a House Plan (Blue Print)

Navigation icons: back, forward, search, and other presentation controls.

Blue Print – House Plan (Class)



Your Dream
House

Class House

Your Dream House

- With the House Plan – Blue Print (Class) you can now get a contractor to build your dream house (Object)



Object

Classes and Objects

- An Object is a specific instance (variable) of the data type (class)
- A class is a blue print of an object.



Classes and Objects

- You can make as many houses as you want from a single house plan – Blue Print (Class)



Class House



House1



House2



House3

Objects

Classes – Captures Behaviour as well

- A concept (class) has both properties and behavior.
- We know that dogs and cats behave differently



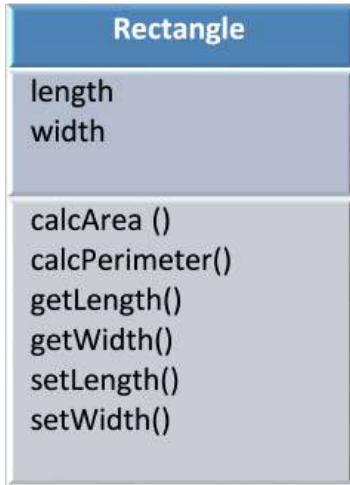
(c) 2017 Monique Snoeck, KU LEUVEN

Example – Behaviour is captured as functions



Grouping properties and functions together is called Encapsulation

Rectangle Class

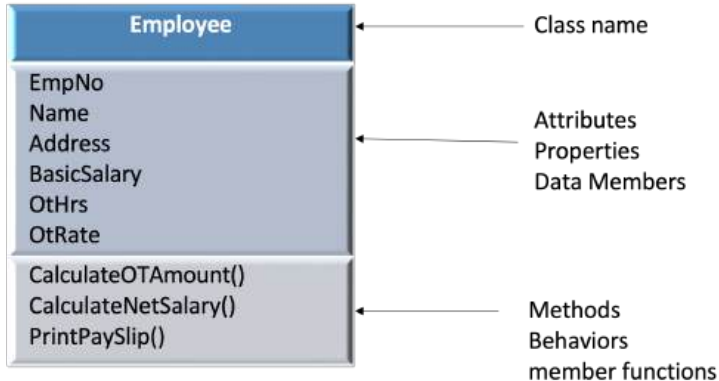


Class Name

Properties
(Attributes)

Functions
(Methods)
(Behaviour)

Employee Class



Restricted Access

- All properties and some functions of a Class have restricted access (private) and can be accessed only through public functions.
- Why is this necessary?
- Let's look at an example.

A jewelry shop





Jewelry can be accessed only through a Sales Person

JewelleryShop Class



Private
(Restricted)

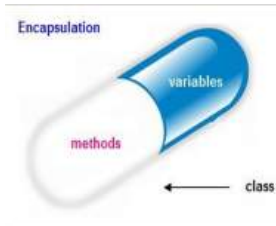
Public
(Salesman
do this)

Information Hiding

- Hide certain information or implementation decision that are internal to the encapsulation structure (class)
- The only way to access an object is through its public interface (public functions)
 - Public – anyone can access / see it
 - Private – no one except the class can see/ use it

Encapsulation

- It is the process of grouping related attributes and methods together, giving a name to the unit and providing an interface (public functions) for outsiders to communicate with the unit.



Encapsulation

- The implementation of the TV is hidden from us. Your TV could be OLED, LCD, Plasma or an old CRT one.
- You can control any such TV using the same commands in your remote (interface).

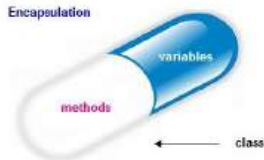


28 / 32

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Encapsulation

- Encapsulation is the process of compartmentalizing the elements of an abstraction that constitute its structure and behavior; encapsulation serves to separate the contractual interface of an abstraction and its implementation.



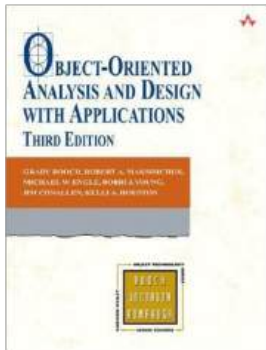
(Reference : Grady Booch, eta (2008), Object Oriented Analysis and Design with Applications 3rd Edition, pg 52)

Interface – Public Functions



You interact with some object through its interfaces.

Your car can be Gasoline, Hybrid or Electric but you drive it the same way.



Chapter 03

Grady Booch (2008), Object-Oriented Analysis and Design with Application, 3rd Edition

Thank You!